

Fed: The balance sheet under Warsh

Rethinking the Fed's footprint under a new Chair



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Fed: The balance sheet under Warsh

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KEY MESSAGES

We expect the Fed to make a gradual, predictable and well-signposted effort to shrink its balance sheet over new Chair Warsh's term.

To achieve this, the Fed must first reduce reserve demand. Changes to supervisory and regulatory requirements are a necessary prerequisite for a smaller balance sheet, in our view.

In our base case, the balance sheet shrinks by ~\$700bn, starting in 2028.

US rates: We see swap spreads falling 3-5bp and term premium rising ~15bp.

US credit: We expect credit spreads to remain tight. Regulatory reform could lead to Banks outperforming.

US equities: We see rotation and dispersion between equity sectors. Like in credit, Bank stocks also have scope for outperformance.

FX: A smaller Fed balance sheet will be USD supportive, but likely with a lag.

History suggests that if Warsh's eventual Quantitative Tightening program is larger, faster or more poorly communicated than we assume, the market reaction can be much more adverse.

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We see the balance sheet shrinking gradually

Economics

Warsh to pursue smaller balance sheet: We think shrinking the central bank's balance sheet will be among Fed Chair Warsh's top priorities. To achieve this will require the Fed to first reform supervisory and regulatory (S&R) requirements, in our view. This will take time, with 2028 being the fastest realistic outlook. We expect balance sheet adjustments to take place gradually and predictably, with changes in related flows announced infrequently, well in advance, and following a reasonably transparent framework.

Gradual and predictable: In our base case, reserve demand could drop as much as ~\$700bn following reforms, with the balance sheet decreasing afterward. We note, however, high uncertainty around these forecasts and will update the base case as we receive new information.

In practice, we think the Fed will monitor the spread between the Tri-Party General Collateral Rate (TGCR) and the Interest on Reserve Balances Rate (IORB) and adjust balance sheet growth based on this: faster if TGCR softens (because of weaker reserve demand) and slower if it firms up. This empirically driven approach would emphasize to the market that market stability is the Fed's most important policy objective.

Aware of tantrum risks: Ample historical experience shows that markets can have outsized, macroeconomically relevant reactions to changes in balance sheet policy – especially the communication around such moves – even before they happen. This occurred in 2013, 2018 and 2019.

Early signs that these outcomes can be avoided on this occasion seem positive. Treasury Secretary Bessent, seemingly aware of such risks, has indicated that “they'll probably sit back, take at least a year to decide what they want to do”. Consequently, in our base case, market moves are modest.

Markets

Rates: We see the rates market reaction from a smaller balance sheet as relatively contained: swap spreads falling 3-5bp across the curve and term premium rising approximately 15bp.

The magnitude of market reaction to Quantitative Tightening (QT) will depend on three factors: the size of balance sheet reduction, the pace of the process and related communication by the Fed. Our base case assumes a modest degree of balance sheet reduction at a gradual pace, with very clear and coordinated communication from the Fed.

Credit: Credit spreads should remain in a tight range. Corporate bonds are likely to remain relatively scarce amid high fiscal deficits and a leaner Fed balance sheet. Plus, high yields can continue to spur demand for credit while limiting supply.

As a smaller Fed balance sheet will likely follow regulatory reform, we see scopes for Banks to outperform.

Equities: We think balance sheet reduction will trigger rotation and dispersion between equity sectors. Regulatory reform could potentially boost RoE and incentivize buybacks from US banks – suggesting room for outperformance.

At the index level, equity performance is more likely to be a function of growth expectations and EPS momentum than the size of the balance sheet or the level of rates.

FX: We expect a smaller balance sheet to support USD valuation. But this takes time to play out – a regression of EURUSD against the relative Fed/ECB balance sheet suggests that the USD benefits from a smaller Fed balance sheet relative to the ECB with about a five-month lag.

Why Warsh wants a smaller balance sheet

Warsh has been critical of a large Fed balance sheet for years. He has contended that running a "bloated" balance sheet distorts market functioning, weakens the central bank's credibility and stokes inflation.

November 2010: *"The prices assigned to Treasury securities--the risk-free rate--are the foundation from which the price of virtually every asset in the world is calculated. As the Fed's balance sheet expands, it becomes more of a price maker than a price taker in the Treasury market. And if market participants come to doubt these prices--or their reliance on these prices proves fleeting--risk premiums across asset classes and geographies could move unexpectedly."*

According to Warsh, persistent Treasury buying gives the impression that the Fed is financing government deficits, which can undermine its institutional credibility.

April 2026: *"A large balance sheet where the Fed owns more outstanding debt than many parts of financial markets...that's fiscal policy in disguise."*

June 2010: *"The Fed's institutional credibility is its most valuable asset, far more consequential to macroeconomic performance than its holdings of long-term Treasury securities or agency securities. That credibility could be meaningfully undermined if we were to take actions that were unlikely to yield clear and significant benefits."*

Going forward, Warsh argues for ["redeploying" the balance sheet](#) in the form of lower rates. He believes such a maneuver would allow the Fed to better achieve its dual mandate of stable inflation and maximum employment.

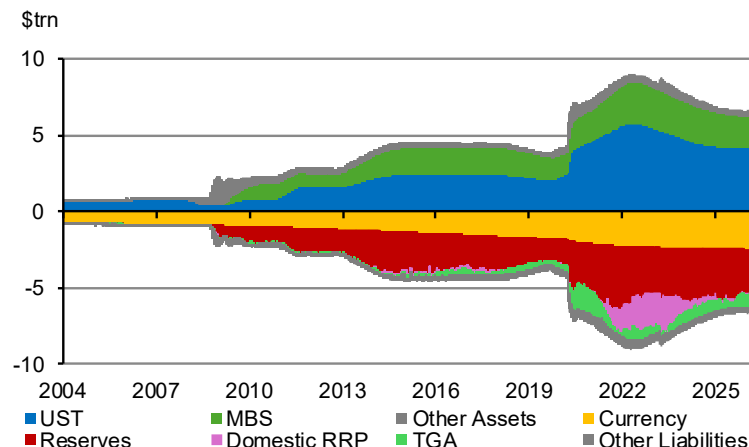
April 2026: *"If the Fed had kept a smaller balance sheet...I think interest rates could be lower...inflation could be better, and the economy could be stronger."*

"If we were to cut rates, then a broader number of people will benefit from it versus quantitative easing, which tends to move through financial assets."

Looking ahead, while Warsh favors balance sheet expansion during a crisis, he has emphasized the importance of communicating a well-defined unwind strategy so markets know the Fed's bond holdings will not become a permanent stimulus tool. In his view, any future decision to expand should be subjected to "strict scrutiny" and justified only when the incremental macroeconomic benefit clearly exceeds the costs. This involves limiting any future debt purchases unless a rigorous, quantitative cost-benefit analysis shows a net positive effect on output or employment.

With the balance sheet much larger now than before the GFC, and with crisis conditions behind us, we see Warsh focusing much of his time and energy on shrinkage here as soon as possible. The aim is to return the Fed to its traditional rate-setting role from that of continuous market participant. However, the Fed must clear some big hurdles before it can work toward this.

Fig. 1: Composition of the Fed's balance sheet over time



Sources: Federal Reserve, BNP Paribas

Why the Fed can't shrink its balance sheet materially now

Before the GFC, the Fed managed a corridor system for reserves with a low stock, representing less than 0.5% of US bank assets (versus ~12% now). Banks held required reserves and voluntary excess reserves, both earning zero interest, with the Fed adjusting reserve supply through daily repo operations to keep the effective fed funds rate (EFFR) near its target (**slide 19**).

After the introduction of quantitative easing (QE), the Fed transitioned to a floor system in 2008, amid an environment of more restrictive capital and liquidity requirements. While it was the Fed that initially supplied the market with an excess of reserves, the system is currently set up such that banks depend on them. Case in point: two sets of balance sheet reduction programs have shown that once reserves (and RRP balances) fall below 11% of bank assets, severe money market volatility ensues, limiting the Fed's ability to materially shrink its balance sheet (**slide 20**). Reversing this reserve demand is therefore key to balance sheet reduction.

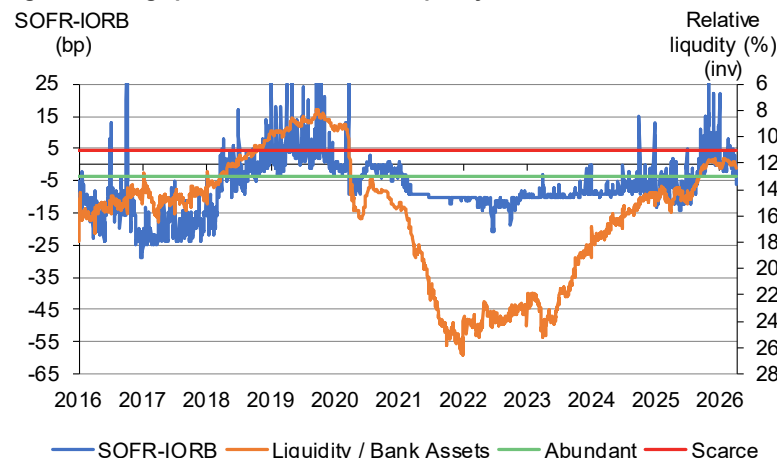
The Fed now faces a "trilemma" by which it can achieve only two of the following three goals at once: a reduced balance sheet, low volatility of short-term rates and limited market intervention. However, we see a path for the Fed to address all three of these hurdles with one effort, namely through S&R reform. More specifically, S&R reform that would reduce: 1) Banks' high-quality liquid asset (HQLA) holdings and 2) structural reserve demand (**slide 21**).

Post-GFC regulations such as the liquidity coverage ratio (LCR) and internal liquidity stress tests (ILST) have resulted in large HQLA holdings by banks. The primary assets in demand are reserves (along with Treasuries), which sit at the very top of the HQLA hierarchy (with no haircuts). These regulations require a bank to maintain an HQLA stock sufficient to meet expected net cash outflows over 30 days, set at 100% for large banks. However, banks typically hold precautionary buffers to avoid triggering supervisory penalties during stress periods (**slide 22**). Encouraging banks to reduce this buffer would go a long way to lower reserve demand and thus shrink the Fed's balance sheet.

But in addition to large HQLA holdings, there are other factors in play that have resulted in a large reliance on reserves. Reducing this reliance would also aid in allowing the balance sheet to shrink. Operational frictions and a perceived stigma surrounding usage of the Fed's liquidity facilities (discount window, SRP) have induced a preference toward reserves. Just as a risk-averse retail investor may keep ample extra funds in a checking account to avoid an overdraft fee, banks also keep extra cash parked at the Fed to avoid running an intraday overdraft or borrowing from the central bank (**slide 23**). That reduces demand in other markets, such as Treasuries and US agency debt, which are supposed to be considered safe and easily convertible into cash in financing markets. We think S&R reform that addresses this structural reserve demand will also be needed before the Fed's balance sheet can contract more materially.

On the following page, we lay out S&R reforms that the Fed and its partner agencies could consider and estimate their impact.

Fig. 2: Funding spreads blew out when liquidity fell below 11% of bank assets



Sources: FRBNY, Federal Reserve, Bloomberg, BNP Paribas

Our base case: \$700bn balance sheet reduction

We see the most likely regulatory reforms to achieve a smaller balance sheet as:

1. Counting discount window capacity toward liquidity requirements, stress tests and discretionary supervisory exams, thus reducing both banks' HQLA holdings and the discount window stigma.
2. Introducing central clearing of the Fed's SRP to improve dealer intermediation and make Treasuries, agency debt and agency MBS reliably convertible to cash at scale, even in times of stress.
3. Upgrading agency debt and MBS to Level 1 HQLA to improve demand relative to reserves.

Discount window reform is a necessary precondition for successful liquidity reform, in our view. Two ways to do this would be de-stigmatizing usage of the discount window and allowing non-HQLA collateral pledged for the discount window to count toward LCR calculations. Banks would likely then be more comfortable holding an LCR closer to 100%, compared to the top GSIBs now holding an average LCR close to 120%. With banks subject to the LCR currently holding over \$4trn in HQLA, the reduction of reserve demand could be substantial.

While this note concentrates on the LCR, comparable reforms to internal liquidity stress tests (ILSTs) and resolution-liquidity rules – specifically eliminating the bans and hurdles that keep discount window funding out of the calculations – would also help to push down reserve demand. **We estimate the combined reform to liquidity buffer requirements may reduce reserve demand by approximately \$400bn.**

Central clearing of the SRP, meanwhile, would provide an alternative source for dealers' liquidity needs, allowing them to reduce their massive precautionary reserve buffers. Dealers are currently hesitant to use the SRP given the balance sheet costs involved. But if the SRP were centrally cleared and counterparties could net SRP borrowing out against reverse repo to other firms, we think it would make the SRP more attractive. **We estimate central clearing could reduce structural reserve demand by \$200bn.**

Finally, on agency debt, MBS issued through Fannie Mae and Freddie Mac make up over 70% of the MBS market but are subject to a 15% haircut (Level 2A) to banks' HQLA calculation, making them significantly less attractive than reserves at present.

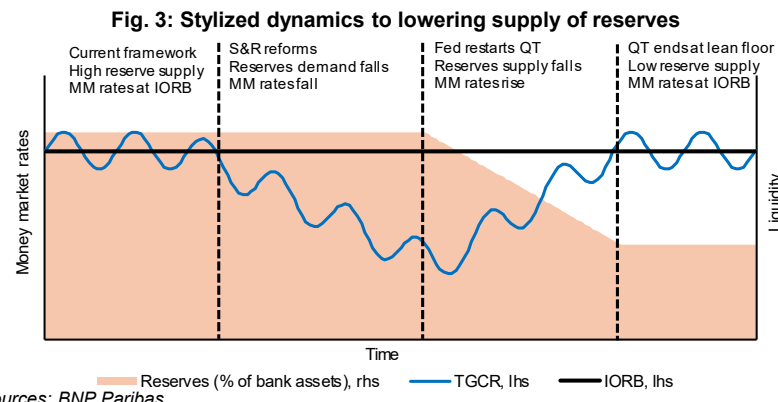
If upgraded to Level 1, MBS issued by these GSEs would have no haircut or cap limiting how much banks can hold. We think this reform would be viewed favorably by the current administration, given the focus on mortgage rates as a barometer of affordability.

The range of impact is difficult to estimate as how banks might shift HQLA holdings based on the above change is unclear. We pencil in a conservative estimate of a **\$100bn reduction in reserve demand from upgrading Fannie and Freddie to Level 1 HQLA**, which assumes that banks increase MBS holdings as a share of total HQLA by just 3%.

Combining these reforms, we expect reserve demand to fall by ~\$700bn.

This could see our estimated ample liquidity range fall 3pp from 11-13% to 8-10%. As this occurs, it should have the effect of putting downward pressure on money market rates (**Figure 3**). Should money market rates persistently settle well below IORB, this would imply that S&R reforms had worked in reducing reserves demand, opening the door for the Fed to safely shrink the balance sheet. Our best guess is that the next round of QT starts 6-12 months after the decline in reserve demand.

We note, however, high uncertainty around these forecasts as Chair Warsh has not yet laid out formal plans on how to reduce the size of the balance sheet. We will update the base case as we receive new information.



A menu of options to shrink the balance sheet

Fig. 4: Hypothetical policy options before restarting quantitative tightening (QT)

Steps	Months	Odds	BS impact (\$bn)	UST impact
Reduce reserves demand				
Publish long-run balance sheet objectives	6	High	----	↔
Adjust liquidity buffer requirements	18	High	400	↓
Centrally clear the SRP	24	High	200	↑
Upgrade agency debt to Level 1 HQLA	12	High	100	↔
Redeploy excess TGA balances into repo*	6	Medium	200*	↑
Reduce the "ample" reserves buffer	3	Medium	200	↓
SLR exclusion of US Treasuries	12	Low	----	↑
Transition to a corridor system				
Reduce FDIC base assessment fees	24	Low	----	
Introduce voluntary reserve targets (VRT)	6	Low	----	

Sources: BNP Paribas

Blank options were unable to be estimated with sufficient precision

* [TGA management reform](#) does not reduce reserves demand, but it does shrink the balance sheet

Our base case of three S&R reforms and \$700bn in balance sheet reduction is likely conservative. Given the extensive menu of options and the multi-year nature of this endeavor, the final roadmap for the Fed's pursuit of a smaller balance sheet may look much different than our expectation.

In a recent [research paper](#), former Fed Governor Miran laid out 15 policy options to shrink the balance sheet. These included a variety of regulatory and operational levers to reduce holdings of reserves and improve substitutes to reserves, with a focus on relaxing liquidity buffer requirements. Overall, his research estimates the combined effect of all 15 policy options could see \$1.2-2.1trn in balance sheet reduction.

Dallas Fed President Logan followed up with a [research note](#) that largely echoed most of Miran's points. She did not provide an estimate for the amount of balance sheet reduction. Instead, the emphasis was on how the Fed could shrink its liabilities through demand-side steps that curb demand for reserves and currency in circulation mixed with supply-side measures such as lowering the TGA buffer or reducing the reserve buffer above the "ample" level.

Ultimately, the Fed will need to assess policy options that would deliver the greatest reduction in the size of its balance sheet with the smallest fiscal cost and lowest systemic risk. This could include a combination of:

- Re-examining bank requirements for liquidity (LCR, ILST) and capital (SLR, GSIB).
- Stigma reduction and operational improvements for the discount window and SRP usage.
- Minimizing growth and volatility in other non-reserve liabilities ([TGA](#), currency in circulation, foreign RRP).
- Introducing a tiered reserve remuneration system such as [voluntary reserve targets \(VRT\)](#) if a scarce-reserve regime is chosen.

The Fed also needs to be exhaustive on educating the public as to how it plans to reduce supply of/demand for reserves in the banking system. A consensus will first need to emerge among policymakers to define the long-run balance sheet goals. This will likely include exhaustive white papers on regulatory reform, de-stigmatization of the ceiling tools, defining "ample" liquidity conditions, and the SOMA portfolio's ultimate composition and duration profile. As such, while Warsh will want to start working on the balance sheet at once, actual shrinkage likely starts only in 2028 at the earliest.

US rates: Market impact from a smaller balance sheet

Fundamentally, the act of shrinking the Fed's balance sheet should lead to cheaper swap spreads (higher Treasury yields relative to swaps) and higher term premiums. The expectation for lower liquidity, elevated Treasury issuance and constrained dealer balance sheets will have an immediate and material impact on the front end of the swap spread curve (**Figure 5**).

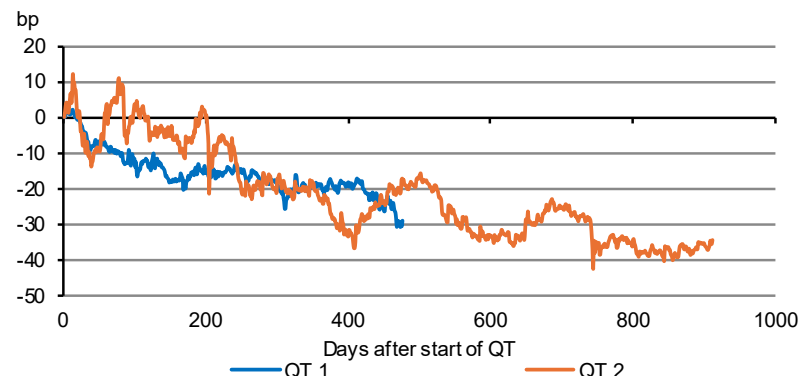
Long-end swap spreads have also become more sensitive to Fed balance sheet headlines over the past couple of years, primarily attributed to rapid growth in duration supply and constraints implied for dealer intermediation in the long run. With several potential or realized amplifiers of US deficits already out there (IEEPA refunds, higher defense spending, stimulus checks), adding QT to the list of items that Treasury needs to fund will only make things harder. Lowering liquidity buffer requirements in isolation may also reduce demand for all HQLA (including Treasuries), not just reserves.

A smaller Fed SOMA portfolio, increased Treasury gross issuance and fiscal concerns should also push term premium higher (**Figure 6**).

Ultimately, however, the magnitude of market reaction to QT will depend on three things: 1) the size of balance sheet reduction, 2) the pace of this reduction, and 3) the Fed's communication and cooperation with Treasury and other agencies to ensure a smooth transition.

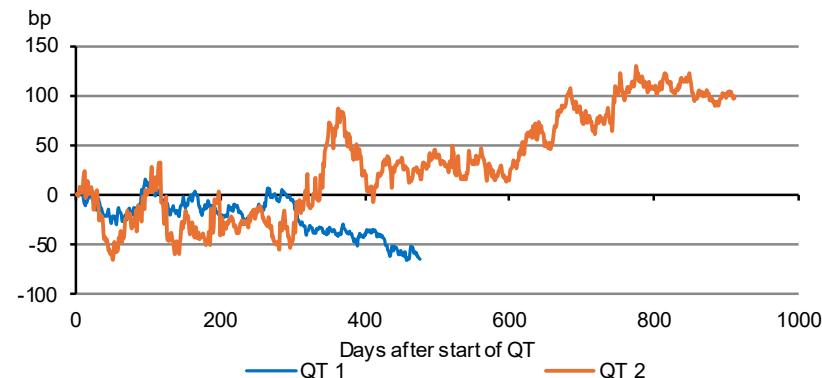
Our base case assumes a modest amount of balance sheet reduction, at a gradual pace, with a very clear and coordinated messaging effort from the Fed. As such, we see the rates market reaction from a smaller balance sheet as relatively contained. However, if the eventual QT program is larger, carried out more quickly, or communicated more poorly than we assume, the market reaction can be much more adverse. On the following page, we attempt to quantify what this would look like.

Fig. 5: 2y swap spreads during past QT episodes



Sources: BNP Paribas

Fig. 6: 10y term premium during past QT episodes



Sources: Bloomberg, BNP Paribas

US rates: Quantifying the market impact

We shock our swap spread and term premium models using a mix of inputs from QT1 and QT2 to arrive at expectations for how the market will react to QT3. **Our base case is for swap spreads to fall 3-5bp and term premium to rise approximately 15bp.** But in a scenario where the Fed starts QT abruptly with poor communication, our models imply front-end spreads could fall as much as 10bp, long-end spreads could come down nearly 20bp, and 10y term premium could rise over 50bp.

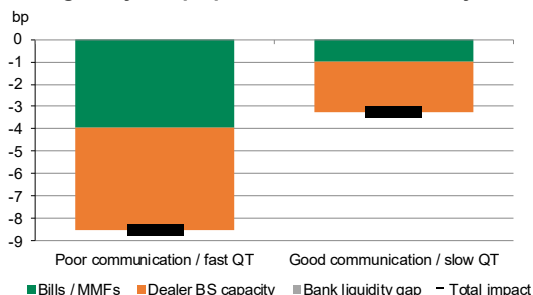
2y swap spreads model: The three key inputs to our model are: 1) the share of T-bill supply soaked up by MMFs, 2) primary dealer UST balance sheet capacity utilization, and 3) banks' liquidity gaps. In the past two QT episodes, bill supply rose for funding, dealer capacity utilization climbed and bank demand for securities fell. To project forward, we assume a similar increase in our bill absorption input, partially trim the dealer capacity jump, and keep our bank demand variable unchanged due to the easing effects of sponsored repo, liquidity

buybacks and regulatory relief (eSLR/GSIB surcharges). In the adverse scenario, we assume a faster jump in bill supply and less primary dealer capacity to absorb Treasuries.

30y swap spreads model: The four key inputs to our model are: 1) the free float of coupon-bearing Treasuries relative to GDP, 2) net asset manager exposure to long-end Treasury futures, 3) a banks liquidity gap variable, and 4) a proxy for equity hedging flows from pension and insurance portfolios. The coupon supply input jumped sharply in both prior QT periods. Asset manager exposure rose as periods of QT also tended to coincide with Fed hikes (flattening curves encouraged longer-duration positions). For the model, we assume a similar rise in coupon supply relative to QT1 and QT2, as the underlying deficit trend is expected to continue and be escalated by QT3. This variable primarily drives the cheaper spreads we expect. In the adverse scenario, we assume an even more aggressive rise in coupon supply, which then implies much cheaper spreads.

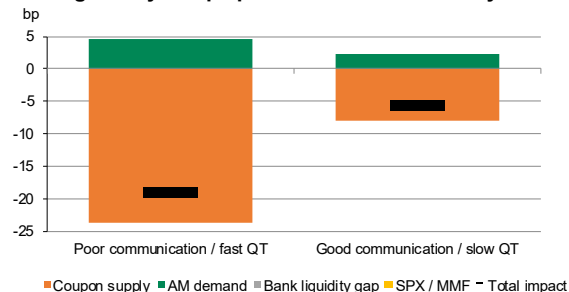
10y term premium model: The five key inputs to our model are: 1) short-term market rates, 2) survey-based long-term inflation expectations, 3) the size of the Fed's SOMA portfolio, 4) gross Treasury issuance, and 5) bank UST holdings. During the past two QT cycles, the 6m T-bill rate rose sharply, long-run inflation expectations fell, the Fed's SOMA portfolio contracted significantly and supply increased. In our modeling of QT3, the SOMA portfolio contraction assumes a \$700bn decline in reserves, similar to the increase in Treasury supply. The bank UST demand and 6m T-bill rate inputs stayed unchanged, as we think the market is unlikely to price in the same level of tightening seen during past QT episodes. We also apply only a modest drop in our projection for LTIE, as the last QT episode saw a decline from extremely high levels. In the adverse outcome, we assume a much larger decline in the Fed's SOMA portfolio, which leads to much higher term premiums. The results of our analysis are presented in **figures 7-9** below.

Fig. 7: 2y swap spreads contribution analysis



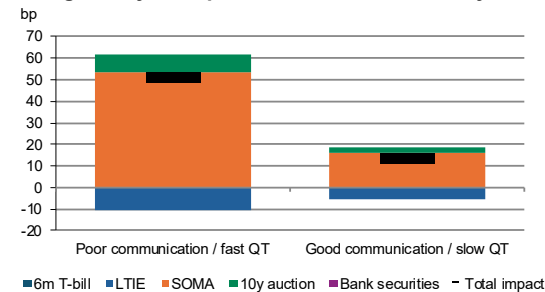
Sources: Federal Reserve, US Treasury, FRBNY, BNP Paribas

Fig. 8: 30y swap spreads contribution analysis



Sources: Federal Reserve, US Treasury, Bloomberg, BNP Paribas

Fig. 9: 10y term premium contribution analysis



Sources: Federal Reserve, US Treasury, Philly Fed, BNP Paribas

US rates: Balance sheet tail risks (i)

The Fed has used its balance sheet more actively, and in bigger size relative to GDP, since the GFC. This change has thrown up many challenges in terms of understanding not only the direct and indirect market impacts, but also the communication around them.

For most of the last 18 years, the Fed's balance sheet has been successfully deployed to help fine-tune financial conditions and ensure steady control of front-end rates. However, there were episodes where the Fed's balance sheet plans created unintended volatility.

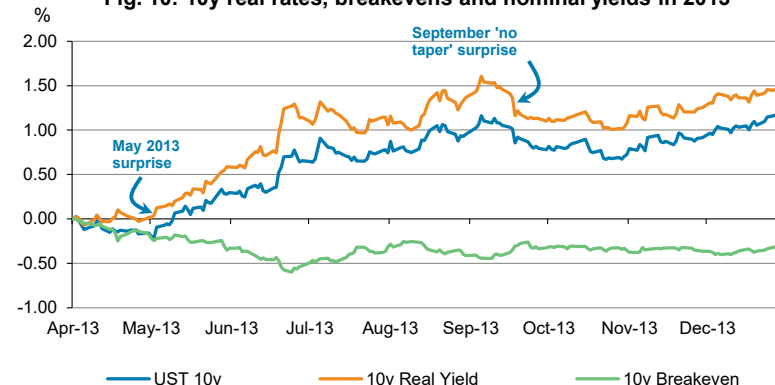
2013's "taper tantrum" was a communication issue, and one that tightened financial conditions sharply. The "mini-tantrum" in late 2018 was similar. The 2019 repo spike was a matter of liquidity misjudgment and drove funding market stress. These were different issues with different consequences, but they all highlight multiple channels where balance sheet policy can go wrong.

If a new balance sheet paradigm emerges under Warsh, some lessons from these past episodes could be helpful in understating the tail risks for US rates investors.

2013 "taper tantrum," a communication challenge: Chair Bernanke communicated at his May 2013 congressional testimony that the Fed was looking to taper its QE program later in the year. Markets were caught off guard and reacted violently relative to the prior calm that had prevailed in a world of zero rates and aggressive balance sheet expansion. 5y and 10y real yields rose nearly 100bp in three months and the 2s10s yield curve steepened.

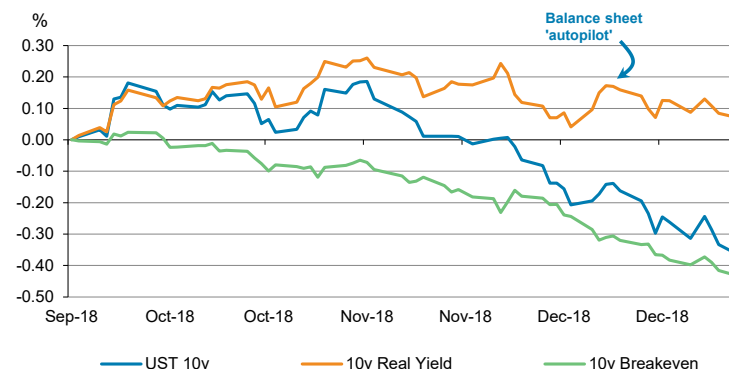
The tightening in financial conditions was sharp enough that the Fed delayed tapering in September 2013, only to cause another volatile swing in US rates. The moves were sharp in both directions, likely tightening financial conditions more abruptly than policymakers preferred. Former Chair Powell referred more than once to the taper tantrum as something that could have been avoided.

Fig. 10: 10y real rates, breakevens and nominal yields in 2013



Sources: Bloomberg, BNP Paribas

Fig. 11: 10y real rates, breakevens and nominal yields in Q4 2018



Sources: Bloomberg, BNP Paribas

US rates: Balance sheet tail risks (ii)

2018 “mini-tantrum,” another communication challenge: In Q4 2018, markets had already been spooked by Powell noting that policy rates were a “long way from neutral”. Poor balance sheet messaging then compounded investor concerns. Specifically, while carrying out the first QT program, Powell said in late December 2018 that balance sheet reduction was on “autopilot”, leading to a large tightening of financial conditions.

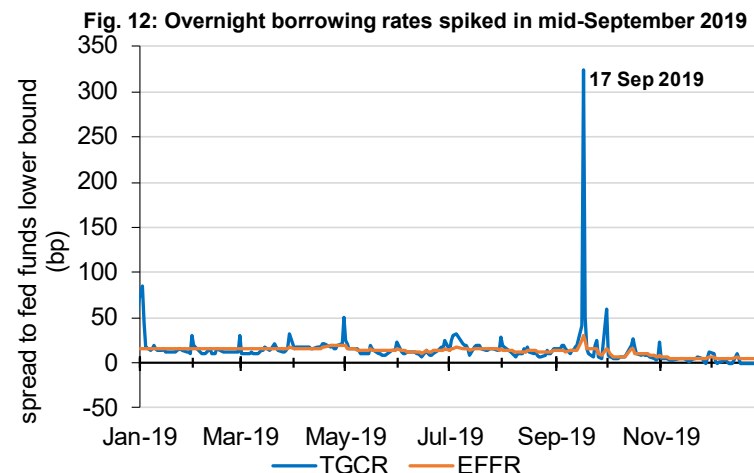
Real rates rose in response, breakevens fell further and equities took a meaningful leg lower. To arrest this abrupt shift, Powell needed to quickly update communications. He scrapped the “autopilot” language, replacing it with assurances that the Fed would not “hesitate to adjust its balance sheet plan” if this interfered with its goals. This again highlighted the importance of a well-planned communication strategy.

2019 repo spike, a balance sheet misjudgment: The Fed’s balance sheet returned to the spotlight in September 2019, not because of any communication issues but rather mainly due to increasing volatility in funding markets, accompanied by a big repo rate spike (**Figure 12**).

The Fed unknowingly reached reserve scarcity and the liquidity shuffle on the mid-September tax date saw a major spike in repo rates, exposing insufficient liquidity in the system. Officials responded quickly by reintroducing TOMOs and announcing a liquidity expansion program through massive reserve management purchases (RMPs) of T-bills (**Figure 13**).

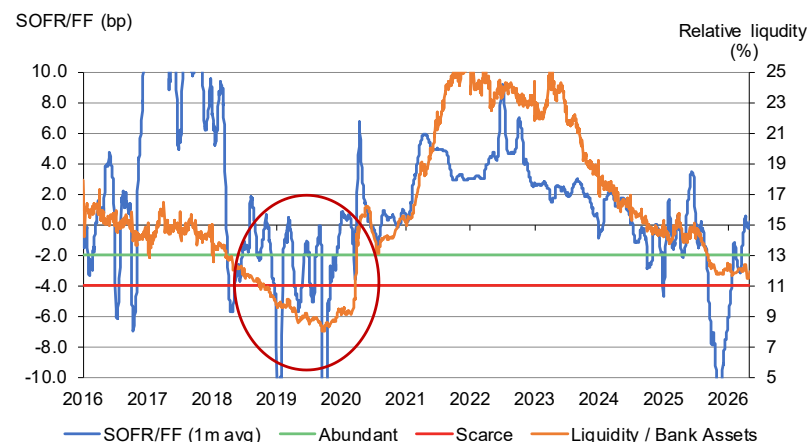
A Warsh-led Fed can change the balance sheet paradigm, but not without risks. The 2019 episode showed that any attempts to move closer to a “corridor system” can create significant volatility in funding markets, with possible further effects. The 2013 and 2018 “tantrums” showed how rates markets and financial conditions can be very sensitive to the Fed’s balance sheet objectives.

Accordingly, while our base case is that any changes to the balance sheet will be well communicated, gradual and predictable, investors should be vigilant for tail risks.



Sources: FRBNY, Bloomberg, BNP Paribas

Fig. 13: Fed unknowingly reached scarce reserve levels in late 2018



Sources: Federal Reserve, Bloomberg, BNP Paribas

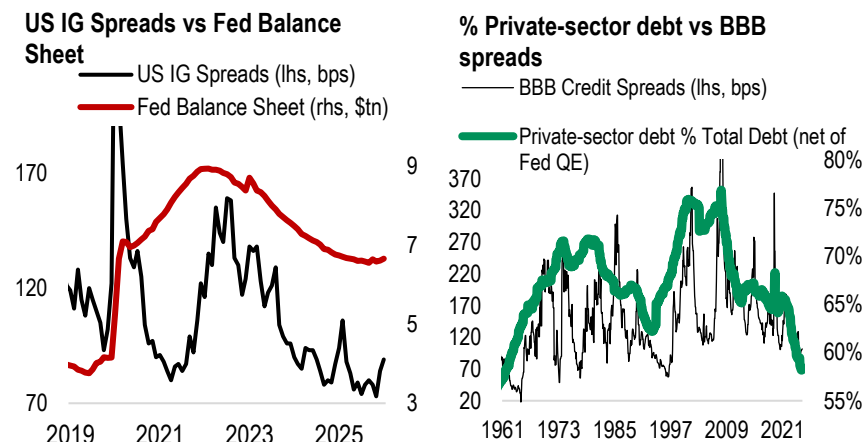
US credit: Treasury abundance, high yields are good for \$ credit

Smaller balance sheet isn't necessarily a credit headwind: Chair Warsh favors a smaller Fed balance sheet, which can lead to higher Treasury yields. A reduction that is gradual and predictable (as we expect) shouldn't be a material risk for credit. Since mid-2022, the balance sheet has shrunk from \$9trn to \$6.5trn, with spreads rallying along the way (**Figure 14, lhs**). We think the credit cycle remains in a benign mid-cycle 'Recovery' (see [here](#)), and a sharp tightening in financial conditions (e.g. another 'taper tantrum' as in 2013) would likely be required to undermine credit cycle dynamics. **If balance sheet reduction is as gradual as we expect, high yields can continue to attract demand for credit while limiting supply – a balance that should keep credit spreads in a tight range, in our view.**

Credit follows the relative scarcity of bonds, not the Fed's balance sheet: Post-pandemic fiscal deficits have 'crowded out' corporate leverage. A fiscal-driven economy can reduce credit risk due to (1) stronger growth and (2) lower corporate debt supply, as high yields raise funding costs. The relative abundance of corporate versus government debt (net of Fed QE) is a secular driver of spreads (**Figure 14, rhs**), our analysis suggests. The relative share of private-sector debt is now at multi-year lows, and a smaller Fed balance sheet would only increase the 'abundance' of Treasuries to be absorbed by investors. Corporate bonds might remain relatively scarce amid high fiscal deficits and a leaner Fed balance sheet, in our view.

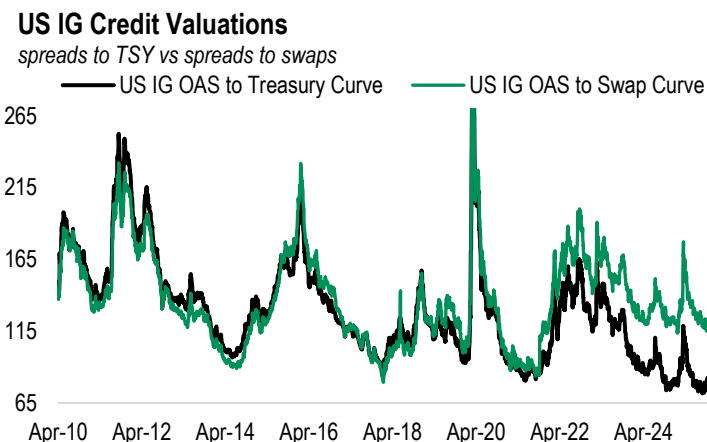
Spreads at tightness versus Treasuries, less extreme versus swaps: The post-Covid 'abundance' of Treasuries is also reflected in the negative level of swap spreads. USD IG spreads remain near this century's tight versus Treasuries, but that partly reflects the 'hedge' (Treasuries) trailing. Credit spreads are less extreme versus swaps (36th percentile over the last 10 years, **Figure 15**), suggesting that multi-year tight \$IG spreads versus Treasuries can be more reflective of Treasuries lagging than of complacency over credit fundamentals, in our view.

Fig. 14: A rally despite QT; spreads follow relative bond scarcity (net of Fed's QE)



Sources: Bloomberg, FRED, BNP Paribas. Note: private-sector debt considers non-financial sectors as defined in Federal Reserve Z.1-Financial Accounts (Flow of Funds components).

Fig. 15: Tight IG spreads: not complacent, just better than its own hedge (TSY)



Sources: BNP Paribas, Bloomberg.

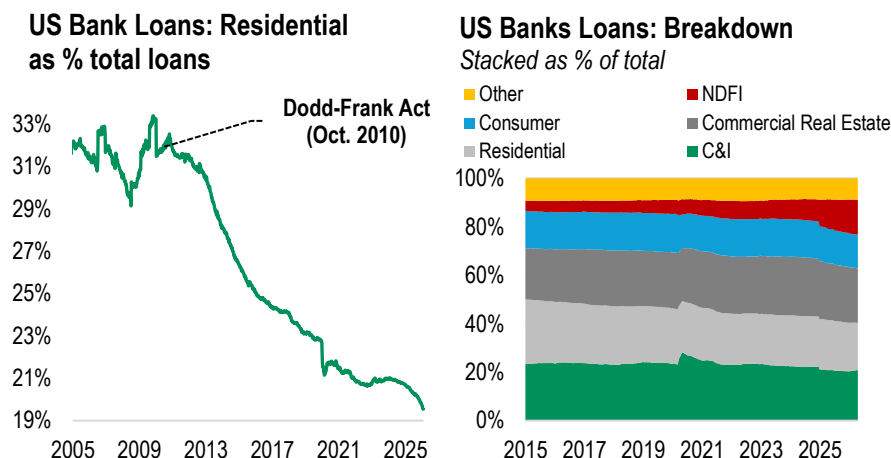
US credit: Regulatory change is a catalyst for Banks to outperform

Reduce regulation-induced reserve demand from US banks: This is the main route to achieve a smaller Fed balance sheet ([Slide 6](#)). US regulatory agencies recently [outlined](#) proposals to reform banking rules. The aim is to deliver some capital and balance sheet relief, consistent with a gradually leaner Fed balance sheet, while potentially incentivizing loan growth over the medium term.

Make mortgage lending great again: Post-GFC regulations made the financial system more resilient, but one side effect has been a migration of mortgage lending away from the regulated banking system (**Figure 16, lhs**). More than two-thirds of all US mortgage lending is now originated by non-banks. Proposed regulation changes aim to incentivize a migration of mortgage origination back into the banking system (along with better treatment of mortgage servicing rights). This would be a headwind for US HY Mortgage Servicers, we think. Meanwhile, US bank loan exposure to Non-Depository Financial Institutions (NDFI, e.g. private credit funds) has increased but remains relatively low (**Figure 16, rhs**). Given robust capital positions, US Banks are unlikely to be a source of systemic risk, in our view.

Looser-than-expected capital requirements can increase RoE: Banks might find themselves with 'excess equity', which we think can incentivize buybacks. For bond supply, the aftermath is more ambiguous: a case for reduced regulation-driven supply needs is possible but would ultimately depend on capital optimization choices and could be countered by longer-term upside risks from faster loans growth. All in all, we think a deregulation move can be bullish for Banks. In our view, **\$IG Banks currently trade with relative appealing valuations versus \$IG Corps, and looser-than-expected capital requirements can drive return on equity higher (this has never recovered to pre-GFC levels, Figure 17), providing a potential catalyst for outperformance.**

Fig. 16: Post-GFC migration of mortgage activity to non-banks



Sources: Bloomberg, BNP Paribas, Federal Reserve. Note: data for Commercial Banks, H.8 Tables.

Fig. 17: Looser than expected capital requirements can drive RoE higher

Large US Banks: Return on Common Equity (%)

LTM, average for large US banks



Sources: BNP Paribas, Bloomberg. Note: average return on equity for an universe of large US banks.

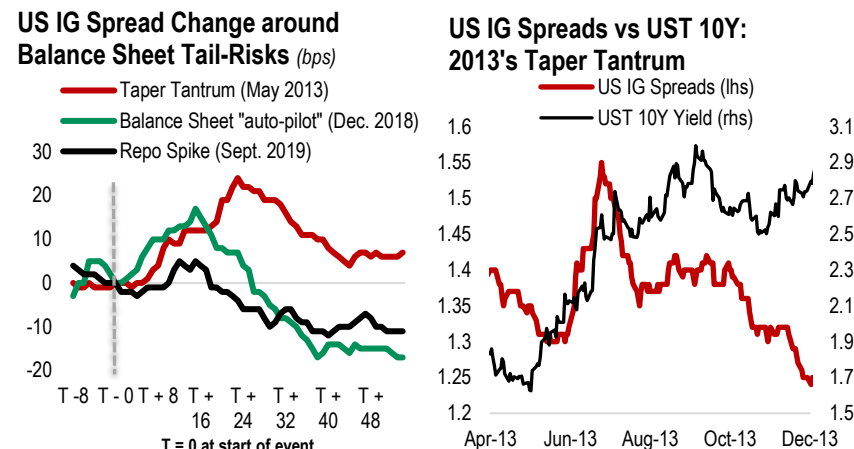
US credit: The risk is in miscommunication (2013, 2018-19 case studies)

Hawkish messaging is the tail risk for \$ credit: Credit can prove resilient during Fed balance sheet shrinkage (e.g. 2023-2025). However, there were episodes when plans in this area had a negative impact on credit spreads. These were mostly about communications that markets took as hawkish surprises. The May 2013 'taper tantrum' was the most negative, with \$IG spreads widening from 130bp to 155bp in the weeks after Chair Bernanke's speech and only fully retracing this by October. More recent fears around balance sheet policy had a smaller impact: the previously noted 'autopilot' episode of December 2018 worsened existing market weakness, but spreads fully retraced the move in less than a month as Chair Powell adjusted his speech. Finally, the September 2019 repo spike was significant for money markets, but \$IG spreads were mostly unchanged (**Figure 18, lhs**).

2013 'taper tantrum' lasted longer than 2018 'autopilot' episode: The usually negative rates-spread correlation can flip to positive during hawkish surprises. Following Chair Bernanke's testimony in May 2013, Treasury yields and IG spreads sold off in tandem with an impact persisting for a few months. However, as rates vol peaked, IG spreads retraced the move even with yields remaining high for longer (**Figure 18, rhs**). Rates vol can be a headwind for credit, but a high-rates regime alongside a well-understood gradual decline in the Fed's balance sheet need not be.

Credit asset allocation playbook for balance sheet tail risks: We looked at relative performance in these three episodes (**Figure 19**). \$HY/\$IG decompression was the norm (less so in 2013), led by \$BBBs trailing versus \$BBBs. Elsewhere, there was some compression in \$BBBs versus \$As (high quality hit by rates sell-off), while the resiliency of \$Bs and \$CCCs reflected how liquidity risk can supersede fundamental risk in these events. The belly lagged the long end, being more exposed to outflows. Across sectors, Banks and Cyclical underperformed. For a balance sheet tail risk event, we would favor \$BBBs, \$Bs and the \$IG long end, as they all proved resilient to past fears around the Fed's balance sheet.

Fig. 18: The bear-case to spreads is a QT miscommunication



Sources: Bloomberg, BNP Paribas. T=0 at 5/22/2013, 12/12/2018 and 9/16/2019, respectively.

Fig. 19: Relative moves as % initial spread levels during risk-events

\$ Credit Asset Allocation / Risk-event	May 2013 Taper Tantrum	Dec. 2018 Balance Sheet 'Auto-Pilot'	Sept. 2019 'Repo Spike'	Average
HY/IG	4.2%	12.3%	13.3%	10%
BBBs/As	-5.3%	0.4%	0.0%	-2%
BBs/BBBs	9.4%	17.4%	20.1%	16%
Bs/BBs	-0.7%	-3.0%	-4.1%	-3%
CCCs/Bs	-6.3%	-6.1%	-6.4%	-6%
Corps/Banks	-10.2%	0.7%	-0.6%	-3%
Cyclical Risk Premium	0.7%	0.5%	1.0%	1%
Belly vs Long-end	3.9%	0.1%	0.4%	1%

Sources: BNP Paribas, Bloomberg. Note: % change in spread ratio of each pair. Moves considered during trough-to-peak movement in \$IG spreads during each risk-event.

US equities: Correlation vs trend

Headline risk: The equity market reaction to balance sheet reduction may initially be driven by communication. Poor communication that coincides with fixed income volatility poses tactical risks to stocks. However, over the medium term, **we think the size of the balance sheet and the level of rates will be back seat drivers for the equity market. Growth expectations and EPS momentum matter most.**

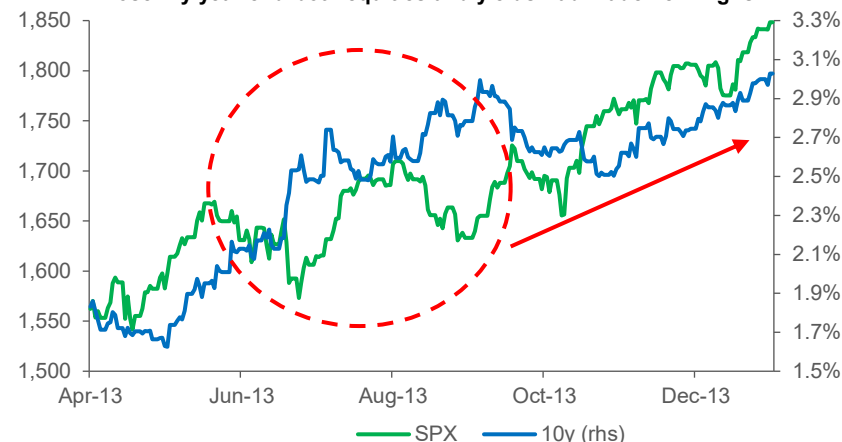
The price action in US equities seen this year already has echoes of 2013. Then, a communication shock saw equities sell-off 5% in four sessions. This move sparked a change in Rate/Equity correlation. US 10y versus SPX realized correlation was +60 going into the taper tantrum. It moved sharply negative in June 2013, and remained flat to negative for the remainder of the year.

After Fed officials walked back more aggressive balance sheet reduction language, it marked the lows for equities. From an intraday low of 1588, the SPX closed the year out at 1874, a 18% rally. This rally came against the backdrop of still higher rates. This reinforces our view that it is more the uncertainty and shock from a balance sheet announcement that could be the trigger for equities than the absolute size of the balance sheet.

YTD we have seen a similar dynamic. Geopolitics in March was a trigger for daily realized correlation between 10y rates and equities to turn sharply negative. However, the point-to-point trend has been both rates and equities higher. YTD 10y rates are up almost 50bp, and SPX and NDX are up 8% and 16% respectively.

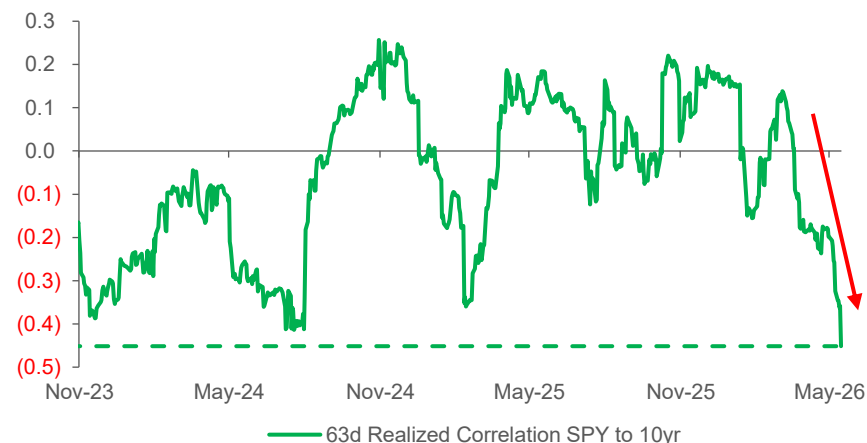
This a playbook we think might remain the right one to use for a balance sheet reduction regime. If there is volatility associated with Fed communication, that could trigger a short and sharp risk-off in equities. Realized correlation for both equities and rates might turn very negative. But as the dust settles, equity delta we see as much more likely to be a function of growth expectations and EPS momentum than the size of the balance sheet, or the level or rates.

Fig. 20: In 2013, initial B/S concerns prompted a 5% SPX pullback while yields rose. By year-end both equities and yields had made new highs



Sources: Bloomberg, BNP Paribas

Fig. 21: Geopolitics in March was a trigger for daily realized correlation between 10yr rates and equities to turn sharply negative. However, the point-to-point trend since has been both rates and equities higher



Sources: Bloomberg, BNP Paribas

US equities: Balance sheet reduction could trigger dispersion

Navigating sector winners and losers: We think a balance sheet reduction could trigger rotation and dispersion between equity sectors. US Banks might also be an interesting idiosyncratic story if regulatory change precedes balance sheet reduction.

A bullish take for Banks: Our credit colleagues ([Slide 13](#)) highlight how regulatory reform presents tailwinds for US banks – potentially boosting RoE and incentivizing buybacks. US Banks are already some of the largest contributors to US buybacks. This comes at time when the largest US mega-caps' capex intentions have investors questioning the longevity of Tech buybacks. **Looser regulation could leave US Banks uniquely placed as a sector with the strongest buyback tailwind in equities.**

Considering the macro scenarios:

Resilient 2013-style macro backdrop – favor cyclical and high-beta: Investors were still reeling from scar tissue from the GFC and the Euro Crisis and had become accustomed to years of ZLB rates and large liquidity injections. Despite being the more notorious 'taper' episode, cyclical, value, and small caps outperformed, driven by a broadly resilient and gradually improving macro backdrop.

The 2018 left-tail risks – favor defensives: The Fed "autopilot" episode during which the economic outlook performed worse than in the other two instances, culminated in the S&P 500 closing just above bear market territory on 26 December 2018. In line with historical expectations, defensive sectors and low-volatility names significantly outperformed the rest of the market.

Muddle-through 2019 – focus on quality: In 2019, following two years of trade tensions under the Trump administration and the fading impact of the Tax Cuts and Jobs Act, investors and markets were left feeling cautious. However, during this episode, the economic outlook was showing signs of stabilization and the Fed had shown it was sensitive to abrupt tightening of financial conditions. This led to subdued, but overall decent performance across equity sectors (ex. Energy). Unsurprisingly, tech, quality, and typical late-cycle winners (HC, utilities, RE) outperformed.

Fig. 22: Macro data likely the main driver of sector dispersion

Name	Price Performance (%)		
	2013	2018	2019
Dates	22-May to 15-Jul	3-Oct to 31-Jan	16-Sep to 31-Oct
S&P 500	1.34	-7.47	1.06
Nasdaq 100	2.51	-9.57	2.82
Russell 2000	5.93	-10.19	-1.55
Tech	0.85	-12.71	3.35
Comm. Svcs.	4.73	-5.87	-0.30
C. Discretionary	4.34	-5.32	-0.81
Financials	3.08	-7.19	1.09
Health Care	1.30	-4.89	3.77
C. Staples	0.02	-0.43	1.17
Materials	-1.38	-8.55	0.03
Industrials	1.46	-9.84	-0.97
Energy	-0.38	-17.69	-8.19
Utilities	-0.63	4.23	2.00
Real Estate	-6.28	5.72	2.01
Value	1.93	-10.22	0.17
Quality	1.46	-9.23	1.69
Growth	0.84	-8.87	1.66
Low Vol	0.72	0.75	1.11
Delta (Δ) in economic backdrop			
Fin conditions (BNP)	-	-	=
Econ activity (BNP)	+	=	-
Econ surprises (BBG)	+	-	=
10y Real Yields	↑	↑	=
Sectorial Takeaways			
Best	Cyclicals, high-beta	Defensives (RE, utils, staples, low vol)	Quality, late-cycle winners
Worst	Defensives, yield sensitive	High-beta, everything else	Cyclicals, late-cycle losers

Sources: Bloomberg, BNP Paribas; *Performance based on ETF price change (%)

FX: Real yields to drive the USD

Academic literature suggests that a larger central bank balance sheet puts downward pressure on the exchange rate. This is because the expansion process often reduces yields. And with less long-term Treasury supply, this contributes to a rebalancing toward duration in the rest of the world, weighing on FX.

A reversal of this trend under Warsh should therefore be USD-supportive. But this takes time to play out. A regression of EURUSD against the relative Fed/ECB balance sheet suggests that **the USD benefits with about a five-month lag from a smaller Fed balance sheet relative to the ECB.**

We do not view this as a tradeable theme for markets in the short term. Indeed, adding balance sheets as a regressor to short-term fair value models actually reduces the statistical significance. What could drive a USD reaction to a balance sheet change is any material impact on the broader macro environment. Looking at historical instances of changes in policy, the 2019 repo spike had little impact on the broader USD, but the currency was more reactive to the wider Fed outlook in 2013 and 2018.

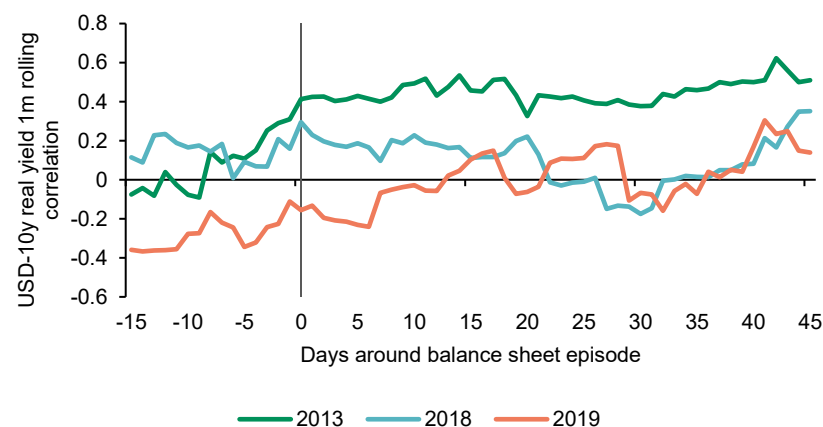
Each of the events was unique, but the common parallel is that the USD was best correlated with 10y real yields (**Figure 23**). Going forward, we suspect that this will continue to drive the FX response.

That said, we see mitigating factors that can limit the extent to which the USD might benefit from a smaller balance sheet. First, if the Fed responds to a smaller balance sheet with rate cuts, buying US fixed income (FX hedged) becomes more attractive. Second, shifting cross-asset correlations may also act as a USD headwind. Higher real yields in 2013 and 2018 were negative for equities, and a stronger USD acted as a natural hedge for foreign investors. But recent risk-off episodes, including Liberation Day in 2025, saw the USD unable to insulate losses when equities came under pressure. Should this positive correlation between stocks and FX reassert itself, foreign investors might have further incentive to hedge their holdings of US equities, weakening the USD despite higher yields.

A history of USD performance over previous balance sheet episodes:

- **2013 'taper tantrum':** Both Chair Bernanke's testimony on 22 May and his speech on 19 June providing a timeline to shrink the balance sheet drove the USD higher alongside real yields, while equities and breakevens fell, consistent with traditional risk-off price action.
- **2018 'mini-tantrum':** The USD rallied from late September to mid-November, driven by EM underperformance after the Fed hiked and Chair Powell used hawkish language. Eventually markets priced a policy mistake and the USD sold off as long positions were unwound into year end.
- **2019 repo spike:** The USD remained in a +/-1% range over the 10 days following the start of this event, consistent with limited spillover to other US assets. The outlook was more driven by US-China trade tensions at the time, indicating that absent spillovers from balance sheet policy to the broader macro environment, the USD should feel little impact.

Fig. 23: USD-10y real yield correlations* during previous balance sheet episodes



*1m rolling. Sources: Bloomberg, BNP Paribas

Tale of two cities → Floor versus corridor

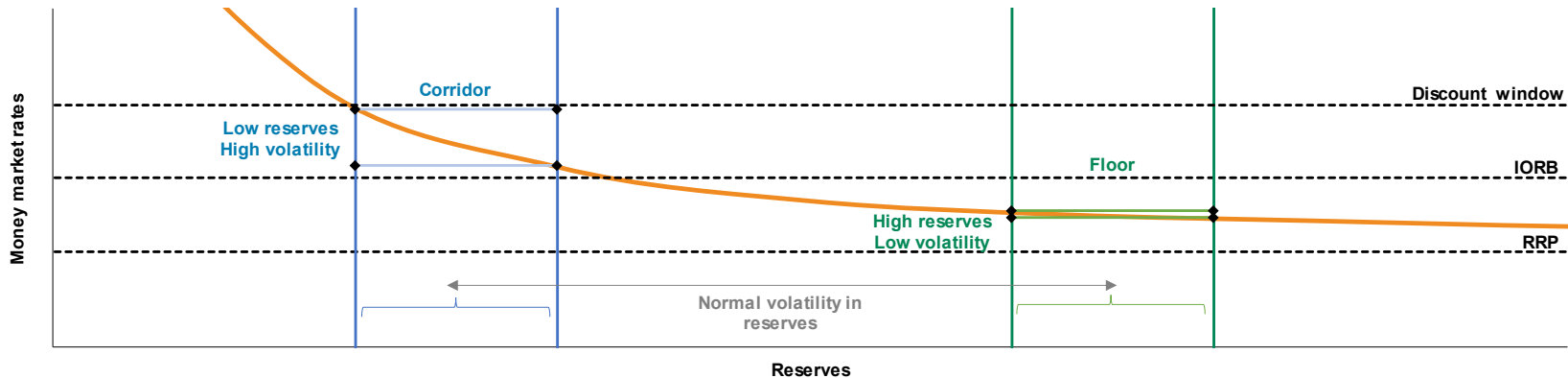
Fig. 24: Characteristics of different balance sheet operating systems

Dimension	Corridor	Floor
Reserve supply	Scarce	Ample (or abundant)
Policy implementation (control lever)	Reserves (supply)	Administered rates (price)
Constellation of rates	Discount window > Market > IORB	IORB > Market > RRP
Interbank lending role	Liquid and critical for reserve distribution	Illiquid and largely irrelevant
Fed operations	Daily temporary open market operations (TOMO)	Structural permanent open market operations (POMO)

Sources: BNP Paribas

IORB = interest on reserve balances, RRP = Fed overnight reverse repo facility award rate

Fig. 25: Reserve demand and money market rate volatility



Sources: FRBNY, BNP Paribas

Calm before the storm → Pre-GFC balance sheet

Prior to the GFC, the Fed operated under a corridor system characterized by a relatively scarce amount of reserves within the banking system. At that time, reserve balances were in the range of double-digit billions and accounted for less than 0.5% of total assets held by US banks.

No Basel-era liquidity buffer requirements such as liquidity coverage ratios (LCR) and internal liquidity stress tests (ILST) were then in place. Instead, banks held reserves at the Fed under two buckets: required and excess reserves.

Required reserves were legally mandated and tied to deposits, while excess reserves were purely voluntary. Both earned zero interest. Required reserves created a baseline demand the Fed could rely on, allowing control of reserve supply to set the fed funds rate to target.

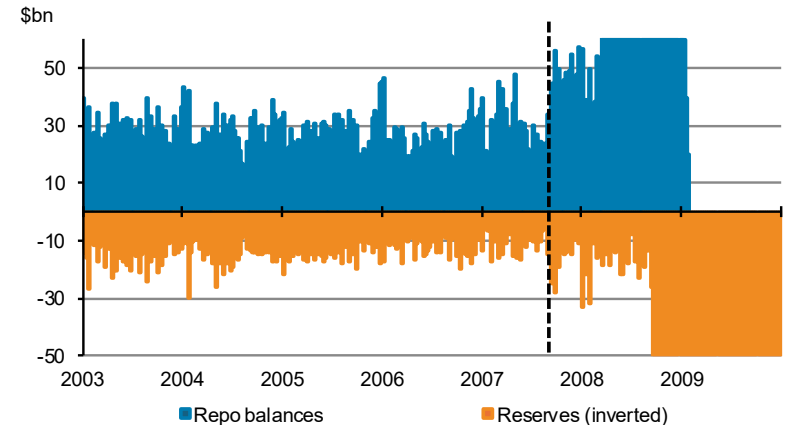
If one bank ran short on reserves, others with excess reserves could capitalize and lend money at the fed funds rate. The Fed's balance sheet goal was simple: to add or drain just enough reserves so that the effective fed funds rate (EFFR) traded close to its target.

Daily changes in other Fed liabilities (TGA, currency in circulation, etc.) and settlement dates would slightly shift supply and demand of reserves. With only mild fluctuations in reserve demand, the Fed used daily repo operations as a precision instrument for keeping the EFFR in check. The repo desk would produce a “reserves forecast” to size these early-morning operations.

Day-to-day dispersion in money markets was then much more acceptable to the Fed (Figure 26). Banks could meet reserve requirements over a two-week maintenance period. If some banks met their requirements early in the maintenance period, they could lend out some of those reserves in the fed funds market if EFFR was trading higher. Therefore, one day's higher EFFR print could be offset by the next day's lower print, which was normal for price discovery (Figure 27).

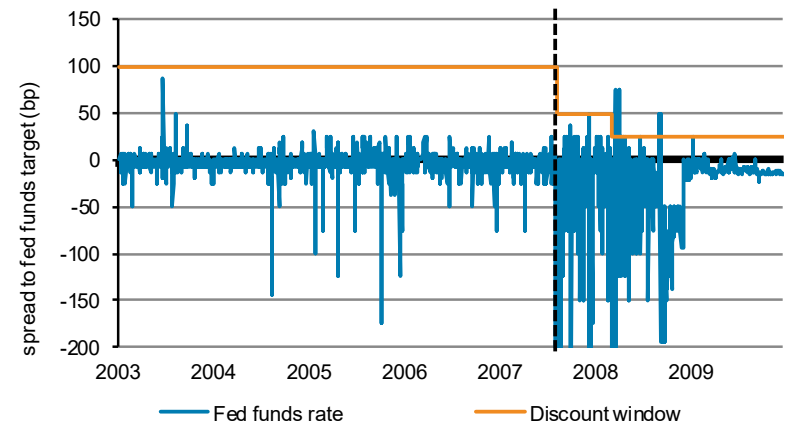
The balance sheet backdrop changed drastically after 2008.

Fig. 26: Repo versus reserves balances pre-GFC



Sources: Federal Reserve, BNP Paribas

Fig. 27: Money market and administered rates versus fed funds target



Sources: ICAP, Federal Reserve, BNP Paribas

C'est la vie → Current balance sheet framework

After 2008, the Fed was forced to transition to a floor system. Fed funds volatility started to pick up following a series of dramatic events:

- 9 August 2007: Headlines of banks limiting withdrawals
- 16 March 2008: Acquisition of Bear Stearns
- 15 September 2008: Lehman Brothers bankruptcy

Banks hoarded reserves and were less willing to lend in the interbank market due to counterparty risk. Financial conditions then tightened dramatically and forced the Fed to cut rates to zero (**Figure 28**).

The Fed also introduced QE by purchasing large amounts of Treasuries and MBS from the private market, flooding the system with liquidity. Once reserves were abundant, the EFFR was approaching the zero lower bound.

In October 2008, the Fed started paying interest on

reserves to keep the EFFR slightly above zero. This allowed for rate hikes or cuts without changing the balance sheet's size. Supply of reserves was no longer a lever to control EFFR, with administered rates becoming the primary tool (**Figure 29**).

The Fed has since remained in an ample (or abundant) reserves regime, engaging in multiple asset purchase programs (including Covid-19) and using administered rates to control money markets.

Fed funds volume largely dried up as domestic banks no longer needed to borrow reserves, and instead just earned interest on reserves (IORB). The fed funds market is now largely made up of US branches of foreign banks (borrowers) and Federal Home Loan banks (lenders) (see [Front-end center: When does fed funds wake up?](#), dated 28 August 2024). Overnight Treasury repo rates have become the cornerstone for money markets due to the exponential growth in Treasury supply that has led to increased reliance on money market funds (MMF) and levered investors to absorb the

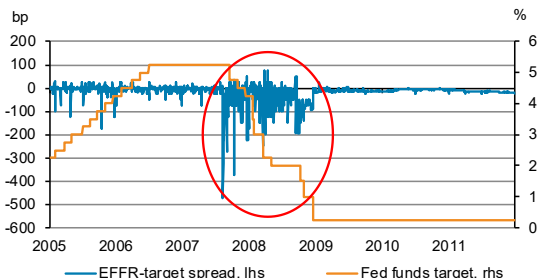
collateral (**Figure 30**).

Besides earning interest on reserves, post-GFC liquidity regulations and discount window stigma have dramatically increased banks' reliance on holding reserves.

In times when liquidity is too abundant and the EFFR is at risk of falling below the fed funds lower bound, the Fed engages in quantitative tightening (QT) by passively letting Treasury/MBS holdings run off its balance sheet. Two rounds of QT have shown that the Fed cannot see reserves (and RRP balances) fall below 11% of bank assets without severe money market volatility (i.e. September 2019).

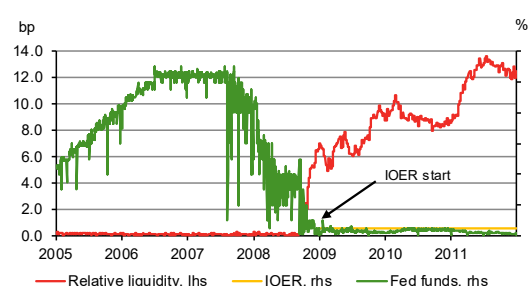
To keep a "neutral" balance sheet in today's environment, the Fed engages in steady reserve management purchases to offset trend growth in other non-reserve liabilities (see [US rates: Fed to buy SO-MAny T-bills](#), dated 3 October 2025).

Fig. 28: Rate cuts start after rising fed funds volatility



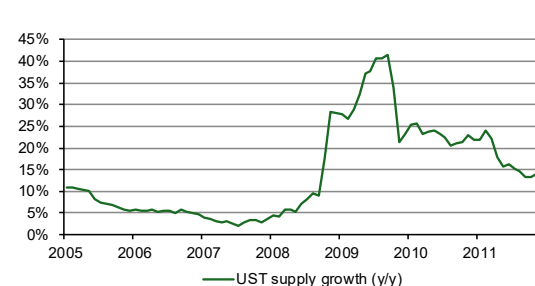
Sources: ICAP, Federal Reserve, BNP Paribas

Fig. 29: Administered rates introduced after QE starts



Sources: ICAP, Federal Reserve, BNP Paribas

Fig. 30: Growth in UST debt supply surges after '08

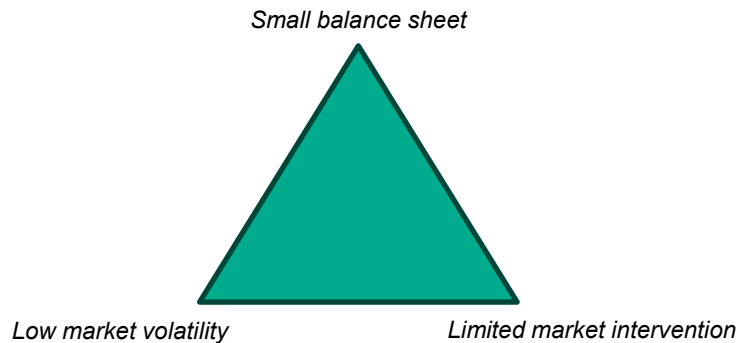


Sources: US Treasury, BNP Paribas

[Sebastian Mauleon](#), US Rates Strategist | BNP Paribas Securities Corp.

Catch-22 → The Fed ‘trilemma’

Fig. 31: The Fed balance sheet ‘trilemma’



Sources: Federal Reserve, BNP Paribas

The Fed previously released a note illustrating the tradeoffs involved in determining the optimal size and dynamics of balance sheet policy, called the [Central Bank Balance-Sheet Trilemma](#). Essentially, it can achieve only two of the following three goals at once: 1) a small balance sheet, 2) low volatility of short-term rates and 3) limited market intervention.

What are the consequences of compromising each of the three goals?

A large balance sheet can help mitigate liquidity shocks and reduce the potential for volatility in money markets, but it also poses duration risk for the Fed and may discourage private-sector activity in money markets.

High money market volatility weakens control over policy rate transmission and reduces liquidity for financing collateral.

Frequent market intervention would curb money market rate volatility in a small balance sheet environment, but would require a periodic “reserves forecast” to determine the amount of reserves the Fed should supply during each maintenance period.

Overall, a larger balance sheet expands the Fed’s structural footprint but also reduces money market volatility and the need for daily market intervention by the central bank. A smaller balance sheet limits the footprint but requires either more active liquidity management by the Fed or higher volatility in money markets.

The appropriate steady-state size of the balance sheet remains an open question (**Figure 32**).

We think a floor system remains the favorable operating system for the Fed. As Governor Waller put it, “*If governments could provide clean, safe drinking water for citizens at zero cost, why would they make it scarce?*”

Fig. 32: Recent balance sheet comments from active Fed officials

Floor		Corridor	
Waller (February 2026)	You don't want banks every night of the day digging around in the couch cushions looking for money ... Scarcity is not the objective in economics. It never has been and it never should be.	Bostic (February 2026)	I think it's doable to go back to scarce reserves.
Powell (June 2025)	If you were to want to go back to scarce reserves [without interest on Fed balances], it would be a long and bumpy and volatile road. I wouldn't recommend that we undertake that road.	Miran (November 2025)	We first have to get the regulations right and ensure that bank balance sheets are flexible enough for an environment with a smaller Federal Reserve footprint.
Logan (November 2023)	As I see it, floor systems have three main benefits. First, rate control in floor systems is more robust ... Second, floor systems continue to effective control rates when the central bank expands its balance sheet ... Third, ... floor systems eliminate what would effectively be a penalty for good liquidity risk management.	Bowman (September 2025)	Over the longer run, my preference is to maintain the smallest balance sheet possible with reserve balances at a level closer to scarce than ample.

Sources: Federal Reserve, BNP Paribas

Safe and sorry → Addressing precautionary reserves

Post-GFC liquidity regulations, such as liquidity coverage ratio (LCR) and internal liquidity stress tests (ILST), made [HQLAs central to bank survival planning](#).

The LCR requires a bank to maintain an HQLA stock that is sufficient to meet expected net cash outflows over 30 days, set at 100% for large banks. Reserves and Treasuries sit at the very top of the HQLA hierarchy and should be used to meet liquidity needs under stress. ILSTs are similar but include slightly longer time horizons (up to one year).

However, regulation requires a bank to notify its supervisor immediately if its LCR falls below 100% (even only briefly) and submit a remediation plan if the LCR is below 100% for three days. As a result, all banks maintain an additional buffer above 100% to mitigate the risk of breaching the regulatory minimum. Certain banks allegedly opted to sell other assets at significantly discounted prices to address liquidity requirements in 2020 rather than use their HQLA stocks, which is the precise scenario the LCR is designed to mitigate. The Bank Policy Institute (BPI) released a comprehensive note on the math behind the [inefficiencies related to the LCR](#) (Figure 33).

There even remains a low degree of substitutability between reserves and Treasuries. They look the same on paper but in operational terms, reserves are more favorable for intraday liquidity needs, settlement mechanics and internal supervisory expectations (i.e. ILST).

One solution could be to make HQLAs usable under times of stress. Regulators would need to create a playbook for banks to gain explicit supervisory permission to draw down these assets in stressed periods without being punished. The BPI recommended lowering LCR requirements during these episodes.

Another solution would be to count prepositioned non-HQLA collateral at the discount window as part of the LCR denominator. Discount window loans can be extended for 90 days, meaning they do not trigger an outflow under the LCR. The BPI recommended taking an ex-ante approach by subtracting [“proven discount window capacity”](#) from the LCR denominator of the LCR. Similar reform would need to apply to ILST requirements as well.

These solutions would free up balance sheet space and allow banks to issue more loans instead of holding large amounts of reserves.

[Sebastian Mauleon](#), US Rates Strategist | BNP Paribas Securities Corp.

Fig. 33: LCR math example in a bank run

$$LCR = \frac{HQLA}{30\text{Day Net Cash Outflow}} \geq 100\%$$

Assume a bank is funded with \$100bn in deposits. The LCR assigns runoff rates for different credit lines. Assume these deposits are subject to a 25% runoff rate. Assume no inflows are expected in the next 30 days.

$$LCR = \frac{HQLA}{\text{Deposits} * \text{Runoff rate}}$$

Scenario 1: LCR at 100%

Pre-bank run period

$$100\% = \frac{\$25}{\$100 * 25\%}$$

Runoff materializes and bank must meet deposit redemptions with HQLA

$$0\% = \frac{(\$25 - \$25)}{(\$100 - \$25) * 25\%}$$

Scenario 2: LCR at 175%

Pre-bank run period

$$175\% = \frac{\$43.75}{\$100 * 25\%}$$

Runoff materializes and bank must meet deposit redemptions with HQLA

$$100\% = \frac{(\$43.75 - \$25)}{(\$100 - \$25) * 25\%}$$

Sources: Bank Policy Institute, BNP Paribas

Plenty of fish → Improving liquidity in other safe assets

Another hurdle to shrinking the Fed's balance sheet is the lack of liquidity in other alternative markets outside of reserves that are meant to be safe, liquid and critical for interbank operations.

Operational frictions and stigma surrounding the usage of the Fed's ceiling tools have induced an over-reliance on reserves. Theoretically, when overnight borrowing rates spike in times of stress, it should be economical for banks/dealers to borrow from ceiling tools. But after 2008, use of the discount window became a public signal of weakness (**Figure 34**). A corridor cannot function with a "leaky" ceiling. If stigma or frictions deter banks/dealers from using the discount window or standing repo program (SRP), then acute money market volatility would become inevitable in a scarce reserves regime. Besides including discount window usage in the LCR, we have discussed other potential solutions to destigmatization before (see [US rates: Hopes and dreams for the SRF](#), dated 20 November 2025)

Moreover, in times of stress, banks/dealers tend to see their balance sheets spike. The supplementary leverage ratio (SLR) subjects them to substantial capital charges on increased reserves, Treasuries and repo balances, forcing them to pull back. While reserves count toward the leverage ratio, banks/dealers would still look to reserves for liquidity survival with intermediation in other markets taking the brunt of the pullback. The Fed has already [lowered eSLR requirements](#) for GSIBs, but excluding Treasuries from the denominator for all banks would improve market liquidity during stress episodes (**Figure 35**).

There have also been considerations to upgrade [Fannie Mae and Freddie Mac MBS](#) to Level 1 HQLA with a zero haircut to boost bank participation in the market. This [Sebastian Mauleon](#), US Rates Strategist | BNP Paribas Securities Corp.

would expand credible liquidity alternatives outside of Treasuries and still reduce concentration in reserves.

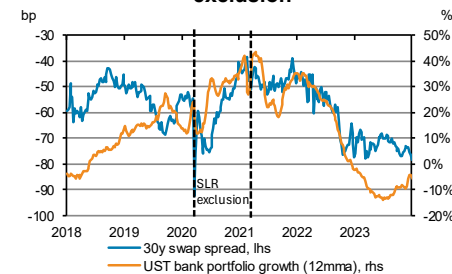
The fed funds market has also largely dried up (**Figure 36**). The FDIC has assessed fees on US banks based on total assets held (rather than just deposits) since 2011. This includes Fed funds and repo borrowed, making IORB arbitrage unprofitable for banks due to the increased balance sheet cost. US branches of foreign banks (FBO) do not face this extra cost. The market now consists of an FBO arbitrage trade to borrow from FHLBs at the EFFR and lend at IORB. Relaxing FDIC fees would bring domestic banks into the trade and improve fed funds market liquidity. With less arbitrage opportunity, the EFFR would print closer to IORB.

However, in a floor system, the lending pool would still be limited to FHLBs. Therefore, truly revitalizing the fed funds market requires a corridor system, allowing banks with marginal excess reserves to become cash lenders and increase the pool of participants in the market. Stress periods would then see a more robust and stable interbank funding market that would avoid an over-reliance on precautionary reserve balances.

Combining it all in an example, suppose a bank estimates an adverse scenario would induce a \$10bn run on deposits. In today's market, a bank may just hold the \$10bn in reserves now because of regulation, stigmatized ceiling tools and a weak interbank market.

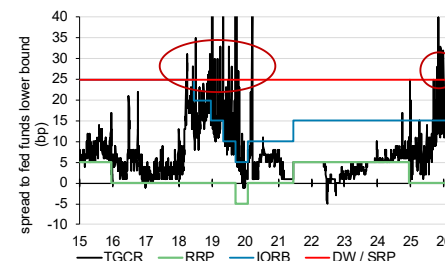
But in a deeper bond market with effective ceiling tools, the bank may be inclined to hold much less in reserves, allocating the remainder to Treasuries and MBS. Additionally, if outflows surpass expectations, a bank could potentially utilize a reliable fed funds market as an alternative instead of being forced to sell assets.

Fig. 34: Treasury bank holdings in SLR exclusion



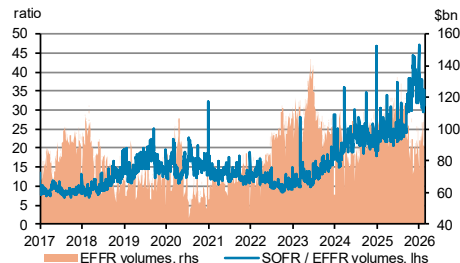
Sources: Federal Reserve, Bloomberg, BNP Paribas

Fig. 35: Ceiling tools do not cap rates well



Sources: FRBNY, BNP Paribas

Fig. 36: Fed funds volumes are low and flat



Sources: FRBNY, Bloomberg, BNP Paribas



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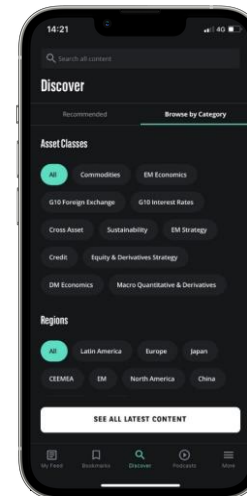
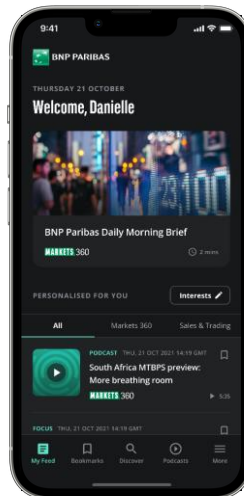
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